

Optimization and Variational Analysis

KKT Conditions, Duality, Stability, and Distributional Optimization

Thanh Nguyen Quoc

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Abstract

This report develops the optimization and variational-analysis foundation for a research program on generative model theory. The main goal is to understand distributional objectives with entropy, KL, and optimal-transport regularization. We review constrained optimization, KKT conditions, Lagrangian duality, mirror-descent ideas, and variational stability of minimizers. These tools provide a mathematical language for diffusion alignment, flow matching, and constrained generative modeling.

1. From Parameter Optimization to Distributional Optimization

Standard deep learning is often written as a parameter optimization problem:

$$\min_{\theta \in \Theta} \mathcal{L}(\theta).$$

For generative models, however, the parameter θ is important primarily because it induces a probability distribution ρ_θ . This suggests an alternative viewpoint:

$$\min_{\rho \in \mathcal{P}(\mathcal{X})} F(\rho),$$

where F measures data fidelity, regularity, alignment reward, or constraint satisfaction. This viewpoint is useful for theoretical analysis because the target object is the generated distribution itself.

A prototypical problem is

$$\min_{\rho \in \mathcal{P}(\mathcal{X})} F(\rho) = \mathbb{E}_\rho[\ell] + \lambda \text{KL}(\rho \| \rho_0) + \gamma \text{OT}_\varepsilon(\rho, \mu).$$

Here ρ_0 is a reference distribution, μ is the data distribution, ℓ is a cost or negative reward, and OT_ε is an entropy-regularized optimal transport cost.

2. Constrained Distributional Optimization

Many generative-model objectives include constraints. For instance, one may require generated samples to satisfy safety, diversity, energy, fairness, or physical constraints in expectation:

$$\mathbb{E}_\rho[c_j(x)] \leq b_j, \quad j = 1, \dots, m.$$

The constrained problem is

$$\begin{aligned} & \min_{\rho \in \mathcal{P}(\mathcal{X})} F(\rho) \\ & \text{subject to} \quad C_j(\rho) := \mathbb{E}_\rho[c_j] - b_j \leq 0, \quad j = 1, \dots, m. \end{aligned}$$

The Lagrangian is

$$\mathcal{L}(\rho, \alpha) = F(\rho) + \sum_{j=1}^m \alpha_j C_j(\rho), \quad \alpha_j \geq 0.$$

A saddle-point formulation is

$$\inf_{\rho \in \mathcal{P}(\mathcal{X})} \sup_{\alpha \geq 0} \mathcal{L}(\rho, \alpha).$$

This formulation is central for primal-dual algorithms and for KKT-type optimality conditions.

3. KKT Conditions in a Finite-Dimensional Approximation

To build intuition, consider a finite sample space $\mathcal{X} = \{x_1, \dots, x_N\}$. A probability distribution is a vector $p \in \Delta_N$, where

$$\Delta_N = \{p \in \mathbb{R}^N : p_i \geq 0, \sum_{i=1}^N p_i = 1\}.$$

Consider

$$\min_{p \in \Delta_N} \sum_{i=1}^N p_i \ell_i + \lambda \sum_{i=1}^N p_i \log \frac{p_i}{q_i}$$

subject to

$$\sum_{i=1}^N p_i c_{j,i} \leq b_j, \quad j = 1, \dots, m.$$

The Lagrangian is

$$\mathcal{L}(p, \alpha, \tau) = \sum_i p_i \ell_i + \lambda \sum_i p_i \log \frac{p_i}{q_i} + \sum_j \alpha_j \left(\sum_i p_i c_{j,i} - b_j \right) + \tau \left(\sum_i p_i - 1 \right).$$

For an interior solution $p_i > 0$, stationarity gives

$$\ell_i + \lambda \left(1 + \log \frac{p_i}{q_i} \right) + \sum_j \alpha_j c_{j,i} + \tau = 0.$$

Solving for p_i yields the Gibbs form

$$p_i = \frac{q_i \exp \left[-\lambda^{-1} \left(\ell_i + \sum_j \alpha_j c_{j,i} \right) \right]}{Z(\alpha)},$$

where $Z(\alpha)$ is the normalizing constant. This formula shows how KL regularization produces a smooth reweighting of the reference distribution.

4. Duality and Interpretation

The dual variables α_j can be interpreted as prices for constraint violation. If a constraint is inactive, the corresponding multiplier is zero. If a constraint is active, the multiplier is generally positive.

The complementary slackness condition is

$$\alpha_j^* C_j(\rho^*) = 0, \quad j = 1, \dots, m.$$

This condition is highly relevant for constrained generative modeling. For example, if a generated distribution violates a diversity or safety constraint, the dual variable increases and penalizes the next primal update.

5. Mirror Descent and KL Geometry

Euclidean gradient descent is not always natural for probability distributions because the simplex has positivity and normalization constraints. A more suitable method is mirror descent with KL divergence:

$$p_{k+1} = \arg \min_{p \in \Delta_N} \left\{ \langle g_k, p \rangle + \frac{1}{\eta_k} \text{KL}(p \| p_k) \right\}.$$

This update has the closed form

$$p_{k+1,i} \propto p_{k,i} \exp(-\eta_k g_{k,i}).$$

This multiplicative update is stable on the simplex and connects naturally with entropy-regularized distributional optimization.

6. Primal-Dual Algorithmic Template

For constrained distributional optimization, a basic primal-dual template is

$$\rho_{k+1} = \arg \min_{\rho} \left\{ \langle G_k, \rho \rangle + \frac{1}{\eta_k} \text{KL}(\rho \| \rho_k) + \gamma \text{OT}_{\varepsilon}(\rho, \mu_n) \right\},$$

$$\alpha_{k+1} = [\alpha_k + \beta_k C(\rho_{k+1})]_+.$$

The research task is to establish conditions under which

$$\frac{1}{K} \sum_{k=1}^K (F(\rho_k) - F(\rho^*)) \rightarrow 0,$$

and

$$[C_j(\bar{\rho}_K)]_+ \rightarrow 0.$$

7. Variational Stability of Minimizers

Let F_n be an empirical or approximate objective and F be the population objective. The key question is:

$$F_n \rightarrow F \implies \operatorname{argmin} F_n \rightarrow \operatorname{argmin} F?$$

A common strategy is to prove:

1. compactness or tightness of near-minimizers;
2. a liminf inequality;
3. existence of recovery sequences;
4. equicoercivity or tightness of the objective family.

In Gamma-convergence terminology, one proves that F_n Gamma-converges to F . If the minimizers are precompact and the limit problem has a unique minimizer, then minimizers of F_n converge to the minimizer of F .

8. Research-Level Proposition Template

A typical theorem for this research direction may have the following form.

Proposition. Let $\mathcal{X}$ be compact, let $\ell_n \rightarrow \ell$ uniformly, and let ρ_0 have full support. Define

$$F_n(\rho) = \mathbb{E}_\rho[\ell_n] + \lambda \text{KL}(\rho \| \rho_0), \quad F(\rho) = \mathbb{E}_\rho[\ell] + \lambda \text{KL}(\rho \| \rho_0).$$

If $\lambda > 0$, then F_n converges to F uniformly on KL-bounded subsets. Under uniqueness of the minimizer of F , every sequence of minimizers $\rho_n^* \in \text{argmin } F_n$ converges to ρ^* in the topology induced by the chosen divergence or weak convergence.

This proposition is only a template. A complete proof requires specifying topology, compactness, and continuity assumptions.

9. Implications for Generative Models

Optimization and variational analysis clarify several issues in generative modeling:

- KL regularization stabilizes distributional alignment.
- KKT multipliers quantify constraint trade-offs.
- Mirror descent gives natural algorithms on probability spaces.
- Variational convergence explains why empirical or minibatch objectives may approximate population objectives.
- Stability theory explains robustness under data perturbation, noisy rewards, and discretization.

10. Exercises and Projects

1. Prove the Gibbs-form solution for a KL-regularized linear objective on the simplex.
2. Derive KKT conditions for a constrained entropy-regularized problem.
3. Implement mirror descent on the simplex.
4. Implement a primal-dual constrained distribution optimization algorithm.
5. Experimentally compare Euclidean projected gradient and KL mirror descent.
6. Write a note explaining how dual variables can be used for constrained generative alignment.

References

1. S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.
2. R. T. Rockafellar and R. J.-B. Wets. *Variational Analysis*. Springer, 1998.
3. G. Dal Maso. *An Introduction to Gamma-Convergence*. Birkhauser, 1993.
4. C. Villani. *Optimal Transport: Old and New*. Springer, 2009.
5. G. Peyre and M. Cuturi. *Computational Optimal Transport*. Foundations and Trends in Machine Learning, 2019.
6. A. Hyvarinen. Estimation of non-normalized statistical models by score matching. *JMLR*, 2005.

7. P. Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 2011.
8. J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 2020.
9. Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. *ICLR*, 2021.
10. Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. *ICLR*, 2023.
11. A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, and Y. Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. 2023/2024.
12. A. Tong, N. Malkin, K. Fatras, L. Atanackovic, Y. Zhang, G. Huguet, G. Wolf, and Y. Bengio. Simulation-free Schrodinger bridges via score and flow matching. 2023/2024.
13. M. Arbel, A. Korba, A. Salim, and A. Gretton. Maximum mean discrepancy gradient flow. 2019.
14. K. Oko, S. Akiyama, and T. Suzuki. Diffusion models are minimax optimal distribution estimators. *ICML*, 2023.
15. T. Kawata, K. Oko, A. Nitanda, and T. Suzuki. Direct distributional optimization for provable alignment of diffusion models. 2025.